



Nguyen Thanh Cong

Intern AI Engineer

congvn308@gmail.com
(+84) 03256810169
github.com/congngocvt308
Ha Noi City, Vietnam

Professional Summary

A highly motivated Mathematics and Informatics student from HUST with a strong algorithmic foundation and data-driven mindset. Proven capability as an autonomous self-learner who rapidly adapts to complex software stacks and architecture frameworks. Thrives under pressure and possesses a resilient debugging attitude to deliver high-performance AI integration pipelines within tight, demanding deadlines.

Education

HaNoi University of Science and Technology - HUST

Sep 2022 – Present

Bachelor of Science in Mathematics and Informatics

GPA: 3.3/4.00

Coursework: OOP, DSA, Relational Databases, Computer Networks, Linear Algebra & Calculus for AI, Probability and Mathematical Statistics.

Projects

Task-Oriented AI Document-to-Quiz Assistant

10/2025 – Present

Full-Stack AI Mobile Developer (Personal Project)

[GitHub Link](#)

- **Architecture:** Developed a decoupled Client-Server pipeline parsing multi-format math documents into structured quiz sets with precise LaTeX rendering and topic extraction.
- **Frontend (Mobile):** Built a native Android client using Jetpack Compose (MVVM) with a RAM-based pre-processing compression engine (~123KB/page) that cut upload latency by 75%.
- **Backend & AI Integration:** Engineered a Node.js/Express server (Render Cloud) using single-turn dual-task prompts with Gemini 2.5 Flash ($T = 0.0$) to combine OCR and topic generation.
- **Concurrency & Streaming:** Mitigated 429 errors using `p-limit` throttling and Exponential Backoff; integrated Server-Sent Events (SSE) for token streaming and progressive LaTeX rendering (<10s).

AI Edge Object Detection Pipeline (YOLO26 & Docker)

06/2026 – Present

AI Edge Engineer (Personal Project)

[GitHub Link](#)

- **Pipeline Augmentation:** Resolved data imbalance via targeted Roboflow augmentation (1,273 public + self-collected images).
- **Model Training & Metrics:** Fine-tuned YOLO26n (100 epochs, T4 GPU); achieved mAP50 77.7% / mAP50-95 61.7%.
- **Targeted Class Recovery:** Lifted door mAP50 0%→97.6% and mouse to 96.2% via oversampling.
- **ONNX Optimization:** Exported to ONNX, reduced CPU inference latency by 43%.
- **Containerized Deployment:** Dockerized with python-slim base, reduced size by 75%.
- **Techs:** YOLO26n, PyTorch, ONNX Runtime, Docker, Roboflow, OpenCV, Python, Google Colab.

Skills

- **AI/ML & LLMs:** Prompt Engineering, RAG, Math OCR, PyTorch, Scikit-learn, Pandas
- **Programming & Frameworks:** Python, C/C++, Kotlin, Jetpack Compose
- **Databases & Tools:** MySQL, SQLite (Room DB), Git, Server-Sent Events (SSE)